

Qns	Answer	Marks
7(c)(ii)1.	The 3 transitions for the 3 bright lines observed are: $E_{3 \rightarrow 2}$: 1.89 eV $E_{4 \rightarrow 2}$: 2.55 eV $E_{5 \rightarrow 2}$: 2.86 eV Hence energy given to the electron must be from $E_{1 \rightarrow 5}$ so that the 3 transitions within the visible light spectrum can take place. $E_{1 \rightarrow 5} = 13.6 - 0.54 = 13.1 \text{ eV}$ Hence $V = 13.1 \text{ V}$	1 1
7(c)(ii)2.	$\lambda = \frac{hc}{E_{5 \rightarrow 2}}$ $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(2.86)(1.60 \times 10^{-19})}$ $= 4.35 \times 10^{-7} \text{ m}$	1

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$2(a)(i)$	when two or more waves meet and overlap, resultant displacement is	